Code No: 132AD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, September - 2023
COMPUTER PROGRAMMING IN C
(Common to EEE, ECE, CSE, IT, ETM)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	How real numbers and integers are stored.	[2]
b)	What is meant by time sharing environment?	[3]
c)	Explain about scope rules.	[2]
d)	What are the limitations of recursion?	[3]
e)	Write about Pointer Arithmetic with an example.	[2]
f)	List the substrings that can be formed from the string "JNTUH".	[3]
g)	Write short note on Enumerations.	[2]
h)	Explain self-referential structures in brief.	[3]
i)	Write about File streams.	[2]
j)	Discuss about fopen() function.	[3]

PART - B

(50 Marks)

2.a) Explain in detail about Selection Statements in C.
b) Write an algorithm to find the greatest number among the three given numbers. [5+5]

OR

3.a) Explain in detail about Computer Languages.
b) Explain the use of 'for' loop, 'do-while' with the help of example program. [5+5]

4.a) Discuss various applications of arrays.
b) Write a program to calculate GCD of two numbers by using functions. [5+5]

OR

5.a) Write a program to implement the binary search by using functions.
b) What is an array? Explain the one-dimensional arrays for Inter Function Communication. [5+5]

6.a) Discuss about String Input / Output Functions with example programs.
b) Explain in detail applications of pointers. [5+5]

OR

7.a) What is meant by arrays of strings? When it will be used? Explain with a 'C' program.
b) Write a program for implementation of pointers to pointers. [5+5]

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8.a) Explain structures and functions with an example program.
b) Write a program to demonstrate command line arguments. [5+5]

OR

9.a) Explain the Enumerated data types with an example.
b) Write a C program to implement unions. [5+5]

10.a) Explain the different types of files are used in C.
b) Write a C program to implement feof() and ferror() functions. [5+5]

OR

11.a) Compare formatting input/output functions with scanf() and printf().
b) Write a program to copy the content of one file to another file. [5+5]

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